SPANNING TREES OF SMALL DIAMETER AND THE RANDIĆ INDEX: A REDUCTION OF FAJTLOWICZ'S RADIUS CONJECTURE TO 2-CONNECTED CORES, AND THE CASE OF MAXIMUM DEGREE AT MOST FIVE

ABSTRACT. Fajtlowicz conjectured in 1988 that the Randić index $R(G) = \sum_{uv \in E(G)} (d_u d_v)^{-1/2}$ of every connected graph G satisfies $R(G) \geq \mathrm{rad}(G) - 1$. We study the stronger inequality $R'(G) \geq \rho(G)$, where $R'(G) = \sum_{uv \in E(G)} 1/\max(d_u, d_v)$ is the modified Randić index of Dvořák, Lidický and Škrekovski, and $\rho(G)$ is the *absolute radius* of G, i.e. half the minimum diameter of a spanning tree. Since $R \geq R'$ and $\rho \geq \max\{\mathrm{rad} - \frac{1}{2},\ \mathrm{diam}/2\}$, this inequality would imply Fajtlowicz's conjecture and the theorem $R' \geq \mathrm{diam}/2$ of Dvořák, Lidický and Škrekovski at the same time; it is tight for all paths with at least three vertices. We develop an exact calculus for R' and ρ under deletion of edges and pendant vertices and under splitting at cut vertices, and show that a minimal counterexample is a 2-connected core carrying pendant vertices under strong local constraints. As consequences we prove: (i) $R'(G) \geq \rho(G)$ for every connected graph of maximum degree at most four, by a short argument (this strengthens the theorem $R' \geq \mathrm{rad} - \frac{1}{2}$ of Cygan, Pilipczuk and Škrekovski for chemical graphs); (ii) the same for maximum degree at most five, with a finite computer check; in particular Fajtlowicz's conjecture holds for all graphs of maximum degree at most five; (iii) the same for every graph whose blocks have at most eleven vertices or cyclomatic number at most eight. In general, $R' \geq \rho$ is reduced to a *core inequality* $\rho(K) \leq R'(K) - 1 + s/3$ for 2-connected graphs K with $s \leq 2$ vertices of degree two and all other degrees at least three. We verify the core inequality for all such K with at most eleven vertices and all such K of cyclomatic number at most eight, prove it for hub-dominated families, and show that a natural rooted strengthening is false. **The core inequality, and with it Fajtlowicz's conjecture for unbounded maximum degree, remains open.**

Status of this report. This is a research report, not a solution of Fajtlowicz's conjecture. Everything labelled Theorem, Proposition, Lemma or Corollary is proved in the text; computer-assisted steps are marked as such and the programs are listed in Appendix A. Statements labelled Conjecture are open.

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1. Introduction

The Randić index of a graph G is $R(G) = \sum_{uv \in E(G)} (d_u d_v)^{-1/2}$, where d_v denotes the degree of v [12]. In 1988 the computer program Graffiti produced the following conjecture.

Conjecture 1.1 (Fajtlowicz [7])**.** *Every connected graph G satisfies $R(G) \geq \mathrm{rad}(G) - 1$.*

Caporossi and Hansen [1] proved $R(T) \geq \mathrm{rad}(T) + \sqrt{2} - \frac{3}{2}$ for trees T and proposed the stronger bound $R(G) \geq \mathrm{rad}(G)$ for all connected graphs except paths with an even number of vertices. Conjecture 1.1 is known for unicyclic, bicyclic and tricyclic graphs [10, 14], for graphs of cyclomatic number at most four (via the harmonic index, Deng, Tang and Zhang [3]), for chemical graphs, i.e. graphs of maximum degree at most four (Cygan, Pilipczuk and Škrekovski [2]), and for cacti (Doig [4]). Dvořák, Lidický and Škrekovski [6] introduced the smaller index

$$R'(G) = \sum_{uv \in E(G)} \frac{1}{\max(d_u, d_v)} \qquad (\leq R(G)),$$

and proved $R'(G) \geq \mathrm{diam}(G)/2$, which gives $R(G) \geq \mathrm{rad}(G)/2$; Yang and Lu [13] proved $R(G) \geq \mathrm{diam}(G)/2 + \sqrt{2} - 1$ for every connected graph, with equality only for paths with at least three vertices. Cygan, Pilipczuk and Škrekovski [2] proved $R'(G) \geq \mathrm{rad}(G) - \frac{1}{2}$ for chemical graphs, and $R(G) \geq \mathrm{rad}(G)$ for chemical graphs other than paths with an even number $n \geq 4$ of vertices, using a structure they called *bags*; they remarked that their proof uses the degree bound in only one step, that the corresponding lemma for general bags would imply Conjecture 1.1, and that the general case had eluded them. The general conjecture is open; see also [5].

1.1. **The problem studied here.** The integer radius is not the right parameter for an exact statement: for the path P_{2k} with $k \geq 2$ one has $R'(P_{2k}) = k - \frac{1}{2} < k = \mathrm{rad}(P_{2k})$. We use instead the *absolute radius* $\rho(G)$, the radius of G regarded as a metric space in which every edge is a segment of length one (Section 2). It satisfies $\mathrm{rad}(G) - \frac{1}{2} \leq \rho(G) \leq \mathrm{rad}(G)$ and $\rho(G) \geq \mathrm{diam}(G)/2$, and $2\rho(G)$ is the minimum diameter of a spanning tree of G (Hassin and Tamir [8]). For every path with $n \geq 3$ vertices, $R'(P_n) = \rho(P_n) = \frac{n-1}{2}$.

Problem 1.2 (spanning-tree form)**.** Does every connected graph G have a spanning tree of diameter at most $2R'(G)$? Equivalently, is $R'(G) \geq \rho(G)$ for every connected graph G?

An affirmative answer would give $R(G) \geq R'(G) \geq \rho(G) \geq \mathrm{rad}(G) - \frac{1}{2}$, i.e. Conjecture 1.1 with $\frac{1}{2}$ to spare, and it would contain the Dvořák–Lidický–Škrekovski theorem because $\rho \geq \mathrm{diam}/2$.

Since $R' \le H$, the harmonic index $H(G) = \sum_{uv} 2/(d_u + d_v)$, it would also give $H(G) \ge \rho(G)$. All connected graphs with at most 11 vertices satisfy $R' \ge \rho$ (Section 9).

1.2. **Results.** Our method is a minimal-counterexample analysis built on three exact tools: an identity for R' when a graph is split at a cut vertex and the two sides are padded with pendant vertices (Lemma 3.3), the matching inequality $\rho(G) \le \rho(G_A) + \rho(G_B) - 1$ for the absolute radius (Lemma 4.3), and a calculus for pendant vertices (Lemmas 3.1 and 4.1). Together they show that a minimal counterexample is a 2-connected *core* K with at most one pendant vertex at each vertex, subject to strong local constraints (Proposition 5.3). A *core* is a 2-connected graph in which exactly $s \le 2$ vertices have degree two and all other vertices have degree at least three.

Theorem 1.3 (reduction; Theorem 5.4)**.** *If every core K satisfies the* core inequality

$$\rho(K) \le R'(K) - 1 + \frac{s}{3}, \tag{1}$$

then $R'(G) \ge \rho(G)$ for every connected graph G, and Conjecture 1.1 holds. The same holds relative to any class of graphs closed under the three operations used in the proof (Definition 5.1).

Theorem 1.4 (Theorems 6.1 and 6.3)**.** *Every connected graph G with maximum degree at most 5 satisfies $R'(G) \ge \rho(G)$. Consequently $R(G) \ge R'(G) \ge \rho(G) \ge \max\{\mathrm{rad}(G) - \frac{1}{2}, \mathrm{diam}(G)/2\}$, and Conjecture 1.1 holds for all graphs of maximum degree at most 5. For maximum degree at most 4 the proof uses no computation.*

Theorem 1.5 (Theorem 7.2)**.** *$R'(G) \ge \rho(G)$ holds for every connected graph each of whose blocks has at most 11 vertices or cyclomatic number at most 8; in particular for every connected graph of cyclomatic number at most 8.*

For maximum degree four, Theorem 1.4 improves the bound $R' \ge \mathrm{rad} - \frac{1}{2}$ of [2] to $R' \ge \rho$; our proof is short because, with the absolute radius, the local constraints on a minimal counterexample leave a single candidate graph (Theorem 6.1). For maximum degree five the constraints bound the size of the core by ten, and the remaining finite set of configurations is checked by computer. For cyclomatic number, Theorem 1.5 extends the bound $\mu \le 4$ of [3] (for Conjecture 1.1) to $\mu \le 8$.

1.3. **What remains open.** For unbounded maximum degree the local constraints no longer bound the core: there are infinite families of configurations that satisfy all of them (Example 8.3). The general case is therefore reduced to the core inequality (1), which we state as Conjecture 8.1. In Section 8 we record what we know about it: it holds for all cores with at most eleven vertices and for all cores of cyclomatic number at most eight; it follows from a weighted max-leaf (connected domination) inequality, which we prove for hub-dominated cores; and its natural strengthening "every vertex of maximum degree is a good centre" is false. We emphasise that *we do not prove Conjecture 1.1 in general.*

1.4. **Organisation.** Section 2 collects facts about the absolute radius; Sections 3 and 4 contain the calculus; Section 5 the structure of minimal counterexamples and the reduction theorem; Section 6 the bounded-degree theorems; Section 7 graphs with small blocks; Section 8 the core inequality; Section 9 the computations.

2. Preliminaries

All graphs are finite and simple. For a graph G we write $d_v = d_G(v)$, $\Delta(G)$ for the maximum degree, and $\mu(G) = |E(G)| - |V(G)| + 1$ for the cyclomatic number of a connected graph. We write $M(G) = R'(G)$ for brevity and call $w_G(uv) = 1/\max(d_u, d_v)$ the *weight* of the edge uv. A *leaf* is a vertex of degree one. Since $\sqrt{d_u d_v} \le \max(d_u, d_v)$ and $2/(d_u + d_v) \ge 1/\max(d_u, d_v)$ we have $M(G) \le R(G)$ and $M(G) \le H(G)$.

Absolute radius. The *metric graph* $\Gamma(G)$ is obtained by identifying every edge with a segment of length one; its points are the vertices and the interior points of edges, and $d(x,y)$ is the length of a shortest path in $\Gamma(G)$. For a point x let $\mathrm{ecc}(x) = \max_{v\in V(G)} d(x,v)$. The *absolute radius* is $\rho(G) = \min_{x\in\Gamma(G)} \mathrm{ecc}(x)$.

Lemma 2.1. *Let G be connected with at least two vertices and radius $r = \mathrm{rad}(G)$.*

(1) The minimum of ecc *on an edge uv is attained at u, at v, or at the midpoint of uv, and* $\mathrm{ecc}(\textit{midpoint}) = \frac{1}{2} + \max_w \min\{d(u,w), d(v,w)\}$.

(2) $\rho(G) \in \{r - \frac{1}{2}, r\}$, and $\rho(G) = r - \frac{1}{2}$ if and only if there is an edge uv such that every vertex is at distance at most $r-1$ from u or from v, i.e. $\max_w \min\{d(u,w), d(v,w)\} \le r - 1$.

(3) $\rho(G) = \frac{1}{2}\min\{\mathrm{diam}(T) : T \text{ a spanning tree of } G\}$.

(4) $\rho(G) \ge \mathrm{diam}(G)/2$.

Proof. (1) Let x be the point of uv at distance $t \in [0,1]$ from u. For a vertex w put $a = d(u,w)$, $b = d(v,w)$, so $|a-b| \le 1$ and $d(x,w) = \min(t+a, 1-t+b)$. If $a = b$ this equals $a + \min(t, 1-t)$; if $a = b-1$ it equals $t + a$; if $a = b+1$ it equals $1 - t + b$. Hence ecc restricted to uv is the maximum of finitely many functions of the forms $c + \min(t, 1-t)$, $c + t$, $c + 1 - t$ with integer c. Such a maximum is minimised at $t = 0$, at $t = 1$, at $t = \frac{1}{2}$, or at a crossing point of an increasing and a decreasing piece, i.e. at $t = (1 + c' - c)/2 \in \{0, \frac{1}{2}, 1\}$. At the midpoint $d(x,w) = \frac{1}{2} + \min(a,b)$. (2) By (1), $\rho = \min\big(r, \min_{uv}(\frac{1}{2} + \max_w \min(d(u,w), d(v,w)))\big)$, and $\mathrm{ecc}(\text{midpoint of } uv) \ge \mathrm{ecc}(u) - \frac{1}{2} \ge r - \frac{1}{2}$. (3) If T is a spanning tree then $d_G \le d_T$ on $\Gamma(T) \subseteq \Gamma(G)$, so $\rho(G) \le \rho(T) = \mathrm{diam}(T)/2$, the last equality because the midpoint of a diametral path of a tree is at distance at most $\mathrm{diam}(T)/2$ from every vertex, while the two ends of the path are $\mathrm{diam}(T)$ apart. Conversely, if $\rho(G)$ is attained at a vertex x, a breadth-first search tree from x has diameter at most 2ρ; if it is attained at the midpoint of uv, take the edge uv together with breadth-first forests from u and v in which every vertex is attached to the nearer of u, v; every vertex is at tree-distance at most $\rho - \frac{1}{2}$ from $\{u,v\}$, so the diameter is at most 2ρ. (4) For vertices a, b and any point x, $d(a,b) \le d(a,x) + d(x,b) \le 2\,\mathrm{ecc}(x)$. □

Part (3) is the classical correspondence between the absolute 1-centre and minimum-diameter spanning trees (Hassin and Tamir [8]); we included the short proof for completeness. A connected dominating set D gives a spanning tree of diameter at most $|D| + 1$ (a spanning tree of $G[D]$ with all other vertices attached to D), hence:

Lemma 2.2. *If G is connected and $\gamma_c(G)$ is its connected domination number, then $\rho(G) \le (\gamma_c(G) + 1)/2$.*

3. The degree calculus of $M = R'$

Lemma 3.1 (leaf deletion)**.** *Let ℓ be a leaf with neighbour p, $d = d_p \ge 2$, and let $\lambda = |\{w \in N(p) \setminus \{\ell\} : d_w \le d - 1\}|$. Then*

$$M(G) - M(G - \ell) = \frac{1}{d} - \frac{\lambda}{d(d-1)} \in \Big[0, \frac{1}{d}\Big].$$

In particular leaf deletion never increases M, and if $d = 2$ and the other neighbour of p is not a leaf, then $M(G) - M(G-\ell) = \frac{1}{2}$.

Proof. Only edges at p change. The edge $p\ell$ had weight $1/d$. An edge pw, $w \ne \ell$, has weight $1/\max(d, d_w)$ before and $1/\max(d-1, d_w)$ after; these differ, by $\frac{1}{d-1} - \frac{1}{d}$, exactly when $d_w \le d-1$. Finally $\lambda \le d - 1$. □

Lemma 3.2 (edge deletion)**.** *Let $uv \in E(G)$ with $d_u, d_v \geq 2$, and let $\lambda_u = |\{w \in N(u) \setminus \{v\} : d_w \leq d_u - 1\}|$, and λ_v symmetrically. Then*

$$M(G - uv) - M(G) = \frac{\lambda_u}{d_u(d_u - 1)} + \frac{\lambda_v}{d_v(d_v - 1)} - \frac{1}{\max(d_u, d_v)}.$$

Proof. As in Lemma 3.1: the edge uv disappears, and an edge uw ($w \neq v$) changes weight, by $\frac{1}{d_u - 1} - \frac{1}{d_u}$, exactly when $d_w \leq d_u - 1$; similarly at v. □

We call an edge e *supportive* if $M(G - e) > M(G)$. Note that λ_u counts a pendant neighbour of u.

Lemma 3.3 (padding identity)**.** *Let v be a cut vertex of G and $V(G) \setminus \{v\} = A \sqcup B$ with $A, B \neq \emptyset$ and no edge between A and B. Put $H_A = G[A \cup \{v\}]$, $H_B = G[B \cup \{v\}]$, $d_A(v) = |N(v) \cap A|$, $d_B(v) = |N(v) \cap B|$, and let G_A (resp. G_B) be H_A with $d_B(v)$ new pendant vertices attached at v (resp. H_B with $d_A(v)$ new pendant vertices at v). Then every vertex of H_A has the same degree in G_A as in G, and*

$$M(G) = M(G_A) + M(G_B) - 1.$$

Proof. In G_A the vertex v has degree $d_A(v) + d_B(v) = d_v$, and the other vertices of H_A keep their degrees, so the edges of H_A keep their weights; the $d_B(v)$ pendant edges have weight $1/d_v$ each. Thus $M(G_A) = \sum_{e \in E(H_A)} w_G(e) + d_B(v)/d_v$, similarly for G_B, and the sum is $M(G) + (d_A(v) + d_B(v))/d_v = M(G) + 1$. □

4. The calculus of the absolute radius

Lemma 4.1 (leaves)**.** *Let G be connected with $n \geq 3$ vertices.*

(1) If ℓ is a leaf, then $\rho(G) - \frac{1}{2} \leq \rho(G - \ell) \leq \rho(G)$.
(2) If L' is a set of leaves whose neighbours are not in L', then $\rho(G - L') \geq \rho(G) - 1$.
(3) If ℓ_1, ℓ_2 are leaves with the same neighbour and G is not a star, then $\rho(G - \ell_1) = \rho(G)$.

Proof. Distances between points of $\Gamma(G - L')$ are the same in G and in $G - L'$, since a pendant edge is never an inner part of a geodesic. (1) Upper bound: let x be an absolute centre of G. If x lies on the edge $p\ell$ at distance t from p, then every vertex $w \neq \ell$ satisfies $d(p, w) = d(x, w) - t$, so $\mathrm{ecc}_{G-\ell}(p) \leq \mathrm{ecc}_G(x)$; otherwise $\mathrm{ecc}_{G-\ell}(x) \leq \mathrm{ecc}_G(x)$. Lower bound: let y be an absolute centre of $G - \ell$ with $\mathrm{ecc}_{G-\ell}(y) = \rho'$; by Lemma 2.1(1) we may assume that y is a vertex or an edge midpoint, so $d(y, p) \in \frac{1}{2}\mathbb{Z}$. If $d(y, p) \leq \rho' - \frac{1}{2}$ then $\mathrm{ecc}_G(y) \leq \rho' + \frac{1}{2}$. Otherwise $d(y, p) = \rho'$; let y' be the point at distance $\frac{1}{2}$ from y on a geodesic from y to p (it exists because $\rho' \geq \frac{1}{2}$, as $G - \ell$ has at least two vertices). Then $d(y', w) \leq \rho' + \frac{1}{2}$ for all $w \neq \ell$ and $d(y', \ell) = \rho' - \frac{1}{2} + 1$. (2) If y is an absolute centre of $G - L'$, every removed leaf is adjacent to a vertex of $G - L'$, so $\mathrm{ecc}_G(y) \leq \mathrm{ecc}_{G-L'}(y) + 1$. (3) By (1) it suffices to show $\rho(G) \leq \rho(G - \ell_1)$. Let y be an absolute centre of $G - \ell_1$. If $y \neq p$ lies on the edge $p\ell_2$, then p is also an absolute centre of $G - \ell_1$: as G is not a star, some vertex $w \notin \{p, \ell_2\}$ exists, every such w has $d(p, w) < d(y, w)$, and $d(p, \ell_2) = 1 < d(y, w)$. So we may assume that $y = p$ or that y does not lie on $p\ell_2$. Then $d(y, \ell_1) = d(y, p) + 1 = d(y, \ell_2)$, hence $\mathrm{ecc}_G(y) = \mathrm{ecc}_{G-\ell_1}(y)$. □

Lemma 4.2 (edge deletion)**.** *If e is not a bridge of G then $\rho(G - e) \geq \rho(G)$.*

Proof. $\Gamma(G - e) \subseteq \Gamma(G)$ and distances in $G - e$ are at least those in G. □

Lemma 4.3 (splitting at a cut vertex)**.** *In the setting of Lemma 3.3, $\rho(G) \leq \rho(G_A) + \rho(G_B) - 1$.*

Proof. Write $\rho_A = \rho(G_A)$, $\rho_B = \rho(G_B)$. Both G_A and G_B have at least three vertices, so $\rho_A, \rho_B \geq 1$. An absolute centre of G_A may be taken in $\Gamma(H_A)$: a point on a pendant edge

at v can be replaced by v, because H_A contains a vertex at distance at least one from v. Let $y_A \in \Gamma(H_A)$ be such a centre and $s_A = d(y_A, v)$. Since G_A has a pendant vertex at v, $s_A \le \rho_A - 1$; moreover $\max_{w\in V(H_A)} d(y_A, w) \le \rho_A$, hence $\alpha_A := \max_{w\in V(H_A)} d(v, w) \le \rho_A + s_A$. Define y_B, s_B, α_B likewise. Distances between points of $\Gamma(H_A)$ are the same in G and in H_A, because v is a cut vertex.

Let P be the path in $\Gamma(G)$ consisting of a geodesic from y_A to v in H_A followed by a geodesic from v to y_B in H_B, and parametrise it by $p \in [-s_A, s_B]$, the signed distance from v (negative on the H_A side). For a point z of P with parameter $p \le 0$ and a vertex w: if $w \in V(H_A)$ then $d(z, w) \le d(z, y_A) + d(y_A, w) \le (s_A + p) + \rho_A$; if $w \in V(H_B)$ then $d(z, w) = -p + d(v, w) \le -p + \rho_B + s_B$. The case $p \ge 0$ is symmetric, so for every p

$$\mathrm{ecc}_G(z(p)) \le \varphi(p) := \max\{\rho_A + s_A + p,\ \rho_B + s_B - p\}.$$

If $p^* = (\rho_B + s_B - \rho_A - s_A)/2 \in [-s_A, s_B]$, then $\varphi(p^*) = \frac{1}{2}(\rho_A + \rho_B + s_A + s_B) \le \rho_A + \rho_B - 1$. If $p^* < -s_A$, then $\rho_B + s_B + s_A < \rho_A$ and $\varphi(-s_A) = \rho_A \le \rho_A + \rho_B - 1$; the case $p^* > s_B$ is symmetric. □

5. Minimal counterexamples and the reduction to cores

Definition 5.1. A class $\mathcal{P}$ of connected graphs is *admissible* if it is closed under (i) deleting an edge that is not a bridge, (ii) passing from G to both padded graphs G_A, G_B of Lemma 3.3, and (iii) deleting a leaf (in graphs with at least three vertices).

Example 5.2. The class of all connected graphs is admissible, and so is the class $\mathcal{P}_D$ of connected graphs of maximum degree at most D (padding preserves all degrees, and (i), (iii) do not increase degrees). More generally, let $\mathcal{Q}$ be a family of graphs that contains K_2 and is closed under taking connected subgraphs. Then the class of connected graphs all of whose blocks lie in $\mathcal{Q}$ is admissible: every block of $G - e$ is a connected subgraph of a block of G; every block of G_A or G_B is a block of G or a pendant edge, i.e. a copy of K_2; and every block of $G - \ell$ is a block of G. For example, for integers N and c the connected graphs with at most N vertices or cyclomatic number at most c form such a family, because neither the number of vertices nor the cyclomatic number increases when one passes to a connected subgraph.

A *core* is a 2-connected graph K that has exactly $s \le 2$ vertices of degree two, all other vertices having degree at least three. Given a graph G and a 2-connected subgraph K such that $G = K + L$, where L is a set of leaves attached to vertices of K, at most one per vertex, we call G a *decorated core*; the vertices of K carrying a leaf are *hosts*, and a host of degree three in G (so of degree two in K) is a *sub*.

Proposition 5.3 (structure of a minimal counterexample)**.** *Let $\mathcal{P}$ be admissible and let $G \in \mathcal{P}$ be a graph with $M(G) < \rho(G)$ that minimises $(\mu(G), |E(G)|)$ lexicographically among such graphs in $\mathcal{P}$. Then $G = K + L$ is a decorated core with $|K| \ge 3$, $K \in \mathcal{P}$, and:*

- *(a) every edge of K is supportive in G;*
- *(b) every vertex of K has a neighbour (in G) of strictly smaller degree; in particular $L \ne \emptyset$;*
- *(c) every $u \in V(K)$ has $d_G(u) \ge 3$; if $d_G(u) = 3$ then u is a sub; subs are pairwise non-adjacent, and every K-neighbour of a sub has degree at least 4;*
- *(d) for every nonempty $L' \subseteq L$, $M(G) - M(G - L') < \rho(G) - \rho(G - L') \le 1$; in particular the number s of subs satisfies $s \le 2$;*
- *(e) every leaf $\ell \in L$ is ρ-critical: $\rho(G - \ell) = \rho(G) - \frac{1}{2}$.*

Proof. Every graph of $\mathcal{P}$ that is smaller than G in the order satisfies $M \ge \rho$. Clearly $n = |V(G)| \ge 3$, since K_1 has $M = \rho = 0$ and K_2 has $M = 1 > \frac{1}{2} = \rho$.

Step 1: every non-bridge edge is supportive. Otherwise $M(G-e) \le M(G)$, $G-e \in \mathcal{P}$ is connected and smaller (μ drops), and $M(G) \ge M(G-e) \ge \rho(G-e) \ge \rho(G)$ by Lemma 4.2.

Step 2: at every cut vertex v, all components of $G-v$ but one are single vertices. Otherwise choose A and B in Lemma 3.3 so that each contains a component with an edge. Then $\mu(G_A) = \mu(H_A) \le \mu(G)$, and if equality holds then $\mu(H_B) = 0$ and $|E(G_A)| = |E(G)| - e(B) < |E(G)|$, where $e(B) \ge 1$ is the number of edges inside B. So G_A, and likewise G_B, is smaller than G and lies in $\mathcal{P}$, and Lemmas 3.3 and 4.3 give $M(G) = M(G_A) + M(G_B) - 1 \ge \rho(G_A) + \rho(G_B) - 1 \ge \rho(G)$.

Step 3: every leaf hangs on a vertex of degree at least three, and no vertex carries two leaves. If a leaf ℓ hangs on p with $d_p = 2$ (and $G \ne P_3$, which is not a counterexample), then by Lemmas 3.1 and 4.1(1), $M(G) = M(G-\ell) + \frac{1}{2} \ge \rho(G-\ell) + \frac{1}{2} \ge \rho(G)$. Stars satisfy $M = 1 = \rho$; if G is not a star and has two leaves at one vertex, Lemmas 3.1 and 4.1(3) give $M(G) \ge M(G-\ell_1) \ge \rho(G-\ell_1) = \rho(G)$.

Step 4: G is a decorated core. If G is a tree, Step 2 says that every vertex has at most one non-leaf neighbour, so G is a star or a double star; stars are excluded, and by Step 3 a double star would have both centres of degree two, i.e. $G = P_4$, which Step 3 also excludes. So G has a cycle and hence a 2-connected block K with $|K| \ge 3$. For a cut vertex $v \in K$ the component of $G-v$ containing $K-v$ has at least two vertices, so by Step 2 all other components of $G-v$ are single vertices. Hence $G = K + L$, where L is a set of leaves attached to K, at most one per vertex (Step 3). All edges of K are non-bridges, hence supportive by Step 1; this is (a). By (iii), $K = G - L \in \mathcal{P}$.

Step 5: (b) and (c). If $u \in V(K)$ has no neighbour of strictly smaller degree, then u carries no leaf and for a K-neighbour w of u Lemma 3.2 gives $M(G-uw) - M(G) = \frac{\lambda_w}{d_w(d_w-1)} - \frac{1}{d_w} \le 0$ (here $\lambda_u = 0$ and $d_w \ge d_u$), so uw is not supportive, contrary to (a). A vertex of minimum degree in K without a leaf would have no neighbour of smaller degree, so $L \ne \emptyset$. A vertex $u \in V(K)$ of degree two in G would have K-degree two and no leaf, and by (b) a neighbour of degree one, which is impossible; so $d_G(u) \ge 3$. A vertex $u \in V(K)$ with $d_u = 3$ must carry a leaf (otherwise it has K-degree three and by (b) a neighbour of degree at most two, contradicting $d_G \ge 3$ on K), so it is a sub. Two subs u, v are never adjacent: we would have $\lambda_u = \lambda_v = 1$ and $M(G-uv) - M(G) = \frac{1}{6} + \frac{1}{6} - \frac{1}{3} = 0$. So every K-neighbour of a sub has degree at least four.

Step 6: (d) and (e). For every non-empty $L' \subseteq L$ the graph $G - L'$ lies in $\mathcal{P}$, is connected and smaller, so $M(G-L') \ge \rho(G-L')$ and, by Lemma 4.1(2), $\rho(G-L') \ge \rho(G) - 1$; since $M(G) < \rho(G)$ we get $M(G) - M(G-L') < \rho(G) - \rho(G-L') \le 1$. Let L_P be the set of leaves of the subs. Removing the leaf of a sub p changes only the weight of the leaf edge ($\frac{1}{3}$) and the weights of the two other edges at p, which stay $1/d_x$ because their other ends x have degree at least four and keep it; subs are pairwise non-adjacent, so $M(G) - M(G-L_P) = s/3$, and hence $s \le 2$. Finally, if a leaf ℓ were not ρ-critical, then $\rho(G-\ell) = \rho(G)$ by Lemma 4.1(1), and $M(G) \ge M(G-\ell) \ge \rho(G-\ell) = \rho(G)$ by Lemma 3.1, a contradiction. □

Theorem 5.4 (reduction to cores)**.** *Let $\mathcal{P}$ be admissible. If every core $K \in \mathcal{P}$ with s vertices of degree two satisfies the core inequality*

$$\rho(K) \le M(K) - 1 + \frac{s}{3}, \tag{2}$$

then $M(G) \ge \rho(G)$ for every $G \in \mathcal{P}$.

Proof. Suppose not and let $G = K + L$ be as in Proposition 5.3. The graph K is 2-connected, has no leaves, and its degree-two vertices are exactly the $s \le 2$ subs, all other vertices having K-degree at least three; thus K is a core in $\mathcal{P}$. By Lemma 3.1, deleting the leaves of non-subs does not increase M, so $M(K) \le M(G - L_P) = M(G) - s/3$. By Lemma 4.1(2), $\rho(G) \le \rho(K) + 1$.

By (2),

$$\rho(G) \le \rho(K) + 1 \le M(K) + \frac{s}{3} \le M(G),$$

a contradiction. □

Applied to the class of all connected graphs, Theorem 5.4 gives Theorem 1.3. The core inequality is tight for K_4 ($s = 0$, $\rho = 1$, $M = 2$) and for no other core with at most 11 vertices (Section 9).

6. Bounded maximum degree

Throughout this section $G = K + L$ is a graph as in Proposition 5.3 for the admissible class $\mathcal{P}_D = \{G : \Delta(G) \le D\}$. For $u \in V(K)$ let $h_u \in \{0, 1\}$ indicate whether u carries a leaf, let a_u be the number of subs adjacent to u, and let f_u be the number of K-neighbours of u of degree 4. For an edge uv of K we write, as in Lemma 3.2, $\lambda_u(v) = |\{w \in N_G(u) \setminus \{v\} : d_w < d_u\}|$; the leaf at u, if any, is counted. By Proposition 5.3(a) and Lemma 3.2, every edge uv of K satisfies

$$\frac{\lambda_u(v)}{d_u(d_u - 1)} + \frac{\lambda_v(u)}{d_v(d_v - 1)} > \frac{1}{\max(d_u, d_v)}. \tag{3}$$

By Proposition 5.3(c), vertices of K of degree three are subs: they carry a leaf, have K-degree two, are pairwise non-adjacent, and for an edge qx from a sub q we have $\lambda_q(x) = 1$.

Theorem 6.1. *Every connected graph G with $\Delta(G) \le 4$ satisfies $R'(G) \ge \rho(G)$.*

Proof. Suppose not and let $G = K + L$ be as in Proposition 5.3 for $\mathcal{P}_4$. Every vertex of K has degree 3 (a sub) or 4, and there are $s \le 2$ subs. For a vertex x of degree 4, $\lambda_x(w) = h_x + a_x - [w \text{ is a sub}]$. Condition (3) becomes:

(i) for an edge qx with q a sub: $\frac{1}{6} + \frac{h_x + a_x - 1}{12} > \frac{1}{4}$, i.e. $h_x + a_x \ge 3$, i.e. $h_x = 1$ and $a_x = 2$ (as $a_x \le s \le 2$);

(ii) for an edge xy between vertices of degree 4: $h_x + a_x + h_y + a_y \ge 4$.

A vertex x of degree 4 without a leaf has, by Proposition 5.3(b), a neighbour of smaller degree, which must be a sub; this contradicts (i). Hence $h_x = 1$ for every vertex of degree four.

If $s \le 1$ then, by (i), no sub has a K-neighbour, so $s = 0$; then every vertex of K has $h = 1$ and $a = 0$, and by (ii) no edge of K is supportive, contradicting Proposition 5.3(a). So $s = 2$, with subs q_1, q_2. By (i) each K-neighbour of a sub is adjacent to both subs; as each sub has exactly two K-neighbours, $N_K(q_1) = N_K(q_2) = \{x_1, x_2\}$. Any other vertex y of K has $h_y = 1$ and $a_y = 0$, so by (ii) all its K-neighbours z satisfy $h_z + a_z \ge 3$, i.e. $z \in \{x_1, x_2\}$; this contradicts $d_K(y) = 3$. Hence $V(K) = \{q_1, q_2, x_1, x_2\}$, $x_1x_2 \in E(K)$ because $d_K(x_i) = 3$, and $K = K_4 - q_1q_2$. Then G is $K_4 - q_1q_2$ with a leaf at each vertex, and a direct computation gives $M(G) = \frac{1}{4} + 4 \cdot \frac{1}{4} + 2 \cdot \frac{1}{4} + 2 \cdot \frac{1}{3} = \frac{29}{12}$ and $\rho(G) = \mathrm{rad}(G) = 2$: indeed $\mathrm{ecc}(x_1) = 2$, every vertex is at distance at least two from some leaf, and for every edge uv some leaf is at distance at least two from both u and v, so Lemma 2.1(2) applies. Thus $M(G) > \rho(G)$, a contradiction. (Alternatively, G violates Proposition 5.3(e): deleting a leaf leaves three leaves at three different vertices, so the same argument gives $\rho(G - \ell) = 2 = \rho(G)$.) □

Corollary 6.2. *For every connected graph G of maximum degree at most four, $R(G) \ge R'(G) \ge \rho(G) \ge \max\{\mathrm{rad}(G) - \frac{1}{2},\ \mathrm{diam}(G)/2\}$.*

This strengthens the inequality $R'(G) \ge \mathrm{rad}(G) - \frac{1}{2}$ of [2, Theorem 1.2] for chemical graphs whenever $\rho(G) = \mathrm{rad}(G)$. (It does not recover the stronger bound $R \ge \mathrm{rad}$ of [2, Theorem 1.3], which uses R rather than R'.)

Theorem 6.3. *Every connected graph G with $\Delta(G) \le 5$ satisfies $R'(G) \ge \rho(G)$. In particular Conjecture 1.1 holds for all connected graphs of maximum degree at most five.*

The proof has a theoretical part, which shows that a minimal counterexample has a core with at most ten vertices, and a finite computation.

Proposition 6.4 (computation)**.** *Let K be a 2-connected graph with $4 \le |K| \le 10$ and $\Delta(K) \le 5$, and let $G = K + L$ be a decorated core with $\Delta(G) \le 5$ that satisfies (a), (b) and (c) of Proposition 5.3 and has at most two subs. Then $M(G) > \rho(G)$. More precisely, there are exactly 128 such configurations $(1, 0, 0, 4, 7, 34, 82$ for $|K| = 4, \dots, 10)$, and the minimum of $M(G) - \rho(G)$ is $\frac{5}{12}$, attained only by $K_4 - e$ with a leaf at every vertex. Moreover, none of the 128 configurations satisfies (e), and all of them except $K_4 - e$ with a leaf at every vertex violate (d) both for $L' = L$ and for the set L' of leaves whose neighbour has degree at most 4 in G.*

Proof. Exhaustive search with the program `cfgcheck.c` ($D = 5$; Appendix A). For each 2-connected K produced by `geng -C -d2 -D5` the decoration is forced except at vertices of K-degree 4: vertices of K-degree 2 or 3 must carry a leaf (by (c)), and vertices of K-degree 5 cannot. All remaining choices are enumerated and conditions (a)–(c) are tested exactly; M and ρ are computed in exact arithmetic, and conditions (d) (for the two sets L' above) and (e) are tested for every configuration. Thus every configuration is excluded in two independent ways: by $M(G) > \rho(G)$, and by the failure of (e), which involves only ρ; all but one are also excluded by (d), which involves only M. □

Proof of Theorem 6.3. Suppose not and let $G = K + L$ be as in Proposition 5.3 for $\mathcal{P}_5$. The vertices of K have degree 3 (subs), 4 or 5; call them of type S, F and P, and write S, F, P for the sets of vertices of these types. Using (3) with $\lambda_x(w) = h_x + a_x - [w \in S]$ for $x \in F$ and $\lambda_y(w) = h_y + a_y + f_y - [w \in S \cup F]$ for $y \in P$, the supportive conditions read:

(i) S–F edge qx: $h_x + a_x \ge 3$, hence $h_x = 1$ and $a_x = 2$;
(ii) S–P edge qy: $h_y + a_y + f_y \ge 2$;
(iii) F–F edge xx': $h_x + a_x + h_{x'} + a_{x'} \ge 4$;
(iv) F–P edge xy: $5(h_x + a_x) + 3(h_y + a_y + f_y - 1) > 12$;
(v) P–P edge yy': $h_y + a_y + f_y + h_{y'} + a_{y'} + f_{y'} \ge 5$.

(For (iv): $\lambda_x/12 + \lambda_y/20 > 1/5$; for (v): $(\lambda_y + \lambda_{y'})/20 > 1/5$; the others as in Theorem 6.1.)

Step 1. As in Theorem 6.1, an F-vertex without a leaf would need a sub as a neighbour, contradicting (i). So every F-vertex carries a leaf, and by (i) it is adjacent to no sub or to both subs. Let F_2 and F_0 be the F-vertices adjacent to two and to no subs. Then $F_2 = \emptyset$ unless $s = 2$, $|F_2| \le 2$, every F_2-vertex has exactly one K-neighbour besides the subs, and by (iii) F_0 is independent.

Step 2 (drop). Let L' consist of the leaves at the subs and at the F-vertices. Removing L' changes the weight $1/\max(d_u, d_v)$ of an edge uv of K only if its end of larger degree loses a leaf. This happens exactly for S–F and F–F edges, whose weight goes from $\frac{1}{4}$ to $\frac{1}{3}$; S–P and F–P edges keep the weight $\frac{1}{5}$. Hence

$$M(G) - M(G - L') = \frac{s}{3} + \frac{|F|}{4} - \frac{2|F_2| + e(F)}{12} \;\ge\; \frac{s}{3} + \frac{|F_0|}{4},$$

because $e(F) \le |F_2|$ (every F–F edge contains an F_2-vertex, which has only one spare K-neighbour). If $L' \ne \emptyset$, Proposition 5.3(d) gives $\frac{s}{3} + \frac{|F_0|}{4} < 1$; if $L' = \emptyset$, then $F_0 = \emptyset$. So $|F_0| \le 3, 2, 1$ for $s = 0, 1, 2$.

Step 3 (every P-vertex has a neighbour of type S or F). Let $J = \sum_{y \in P}(a_y + f_y)$ be the number of edges between P and $S \cup F$. The subs offer $2s$ edge-ends, of which $2|F_2|$ go to F_2;

each F_2-vertex offers one more and each F_0-vertex three, so

$$J \le 2s - |F_2| + 3|F_0| \le 9.$$

Suppose $y \in P$ has $a_y = f_y = 0$. By Proposition 5.3(b) y carries a leaf, so it has four K-neighbours, all of type P, and by (v) each of them, y', has $a_{y'} + f_{y'} \ge 3$. Then $J \ge 12$, a contradiction. Hence every P-vertex has $a_y + f_y \ge 1$, and $|P| \le J$.

Step 4 ($s = 0$ and $s = 1$ are impossible). If $s = 0$ then $F = F_0$. For an F–P edge xy, (iv) with $h_x + a_x = 1$ gives $h_y + f_y \ge 4$, so $f_y = 3 = |F|$. If $F = \emptyset$, Step 3 is violated by any P-vertex. Otherwise each F-vertex has its three K-neighbours in P (as F is independent and there are no subs), each of them adjacent to all of F; so there are exactly three of them, $Y = \{y_1, y_2, y_3\}$, and by Step 3 $P = Y$. Each y_i carries a leaf, so it has K-degree four: three neighbours in F and one in Y. A 1-regular graph on three vertices does not exist.

If $s = 1$, let q be the sub. Its two K-neighbours are P-vertices (an F-neighbour would need $a = 2$). For an F–P edge xy, (iv) gives $h_y + a_y + f_y \ge 4$ with $a_y \le 1$ and $f_y \le |F_0| \le 2$, so $y \in N_K(q)$ and y is adjacent to both F-vertices. Hence an F-vertex would have all three K-neighbours in the two-element set $N_K(q)$; so $F = \emptyset$. Then $h_y + a_y + f_y \le 2$ for all $y \in P$ and (v) never holds: K has no P–P edges, and its only edges are the two edges at q, contradicting 2-connectivity.

Step 5 ($s = 2$). Now $|K| = 2 + |F_2| + |F_0| + |P| \le 2 + |F_2| + |F_0| + (4 - |F_2| + 3|F_0|) = 6 + 4|F_0| \le 10$. So G is one of the configurations of Proposition 6.4, all of which satisfy $M(G) > \rho(G)$ (and none of which satisfies (e)); a contradiction. □

Remark 6.5. We expect that for every fixed D conditions (a)–(e) of Proposition 5.3 bound $|K|$ in terms of D, so that the method of Theorem 6.3 reduces the case $\Delta \le D$ to a finite computation; we have not proved such a bound for any $D \ge 6$. For $D = 6$ there are 4 645 configurations satisfying (a)–(c) with at most two subs and $|K| \le 10$ ($1, 0, 2, 27, 85, 639, 3891$ for $|K| = 4, \dots, 10$); only five of them satisfy $M(G) - M(K) < 1$, none of the 4 645 has all its leaves ρ-critical, and all of them satisfy $M(G) - \rho(G) \ge \frac{5}{12}$. Some dependence on D is unavoidable for the local conditions: for unbounded degree, conditions (a)–(c) together with (d) for $L' = L$ allow arbitrarily large cores (Example 8.3).

7. Graphs with small blocks

Proposition 7.1 (computation)**.** *Every core with at most* 11 *vertices satisfies the core inequality* (2)*. Every core of cyclomatic number at most* 8 *satisfies* (2)*.*

Proof. A core K with n vertices has $2|E(K)| \ge 3n - s \ge 3n - 2$, hence $\mu(K) \ge n/2$; so cores with $\mu \le 8$ have at most 16 vertices and, for $n \ge 12$, $\lceil (3n-2)/2 \rceil \le |E(K)| \le n + 7$. All 2-connected graphs with at most 11 vertices, and all 2-connected graphs with $12 \le n \le 16$ vertices and $\lceil (3n-2)/2 \rceil \le m \le n + 7$ edges, were generated with `geng -C -d2` and those with $s \le 2$ were tested in exact arithmetic (program `corechk.c`). For $(n, m) = (15, 22)$ and $(16, 23)$ the degree sum $2m$ exceeds the minimum $3n - s$ by at most one, so a core has maximum degree at most 4, respectively 3, and the generator was restricted accordingly (`-D4`, `-D3`). Counts and minimum slacks are given in Table 1. There is no violation. □

Theorem 7.2. *Let G be a connected graph each of whose blocks has at most* 11 *vertices or cyclomatic number at most* 8*. Then $R'(G) \ge \rho(G)$. In particular $R'(G) \ge \rho(G)$, and Conjecture 1.1 holds, whenever $\mu(G) \le 8$.*

Proof. The class of such graphs is admissible by Example 5.2, applied to the family $\mathcal{Q}$ of connected graphs with at most 11 vertices or cyclomatic number at most 8. A core belonging to it is a block of itself, so it has at most 11 vertices or cyclomatic number at most 8, and satisfies (2)

by Proposition 7.1. Apply Theorem 5.4. The second statement follows because the cyclomatic numbers of the blocks add up to $\mu(G)$. □

8. The core inequality

Conjecture 8.1 (core inequality)**.** *Every core K with s vertices of degree two satisfies $\rho(K) \le R'(K) - 1 + \frac{s}{3}$.*

By Theorem 5.4, Conjecture 8.1 implies $R' \ge \rho$ for all connected graphs, and hence Conjecture 1.1. We record what we know.

8.1. **Evidence.** Conjecture 8.1 holds for all cores with at most 11 vertices and all cores of cyclomatic number at most 8 (Proposition 7.1); the minimum slack is 0 for K_4 and then $\frac{1}{6}, \frac{1}{6}, \frac{1}{4}, \frac{3}{10}, \frac{1}{3}, \frac{5}{14}, \frac{3}{8}$ for $n = 5, \dots, 11$. Independent random and simulated-annealing searches over cores with up to 40 vertices, including hub-heavy families such as $K_{3,m}$ with extra edges, wheels, books, threshold graphs, blow-ups and necklaces, found no violation; along the families we examined the slack stays bounded away from 0 (for example it equals $\frac{1}{2}$ for all $K_{3,m}$, $m \ge 3$) or grows linearly.

8.2. **A sufficient domination inequality.** Write $I(G) = \sum_{uv \in E(G)} \left|\frac{1}{d_u} - \frac{1}{d_v}\right|$. Since every vertex v satisfies $\sum_{w \sim v} \frac{1}{\max(d_v, d_w)} = 1 - \sum_{w \sim v,\, d_w > d_v} \left(\frac{1}{d_v} - \frac{1}{d_w}\right)$, summing over v gives $|V(G)| - 2M(G) = I(G)$. By Lemma 2.2, Conjecture 8.1 follows from

$$\gamma_c(K) \le 2M(K) - 3 + \tfrac{2s}{3} \qquad \text{equivalently} \qquad L_{\max}(K) \ge I(K) + 3 - \tfrac{2s}{3}, \tag{4}$$

where $L_{\max}$ is the maximum number of leaves of a spanning tree. Inequality (4) is a weighted max-leaf statement: every edge joining vertices of very different degrees must be paid for by leaves. It holds for all cores with at most 10 vertices (with equality only for K_4; the minimum slack for $n = 10$ is $\frac{5}{7}$), and the stronger statement $\gamma_c(G) \le 2M(G) - \max\{2, 3 - \frac{s}{3}\}$ holds for all 9 808 209 connected graphs of minimum degree at least two with 10 vertices and all smaller ones (program `cscheck.c`); it is tight for K_4, for K_4 with one edge subdivided, for $K_4 - e$, for $K_{2,m}$ and for all cycles. When all degrees are 3 or 4, (4) follows for $n \ge 8$ from the theorem of Kleitman and West [9] that $L_{\max} \ge n/4 + 2$ when $\delta \ge 3$: here $s = 0$ and $I = e_{34}/12 \le \min\{n_3/4, n_4/3\}$, which is at most $\frac{n}{7} \le \frac{n}{4} - 1$ for $n \ge 10$, and at most 1 and $\frac{5}{4}$ for $n = 8$ and $n = 9$, where $L_{\max} \ge 4$ and $L_{\max} \ge 5$ respectively. This is the max-leaf counterpart of Theorem 6.1. The following special case covers the near-extremal families $K_{3,m}$.

Proposition 8.2. *Let K be a connected graph of minimum degree at least 3, let A be an independent set of vertices of degree 3, $B = V(K) \setminus A$, let κ be the number of components of $K[B]$, and put $W_B = \sum_{uv \in E(K[B])} 2/\min(d_u, d_v)$. If $W_B \le |B| - \kappa$, then $\gamma_c(K) \le 2M(K) - 3$ and hence $\rho(K) \le M(K) - 1$, i.e. K satisfies the inequalities (4) and (2) with $s = 0$ (here K need not be 2-connected).*

Proof. There are no edges inside A, and every edge between A and B has weight $1/d_b$ for its end $b \in B$ (as $d_a = 3 \le d_b$). Since $2|B| = \sum_{b \in B} \sum_{e \ni b} 2/d_b$, and $\frac{1}{d_u} + \frac{1}{d_v} - \frac{1}{\max(d_u, d_v)} = \frac{1}{\min(d_u, d_v)}$, we get $2M(K) = 2|B| - \sum_{uv \in E(K[B])}(2/d_u + 2/d_v - 2/\max(d_u, d_v)) = 2|B| - W_B$. Note that $|B| \ge 3$: if $A \ne \emptyset$, a vertex of A has three neighbours, all in B, and if $A = \emptyset$ then $|B| = |K| \ge 4$.

Start with a spanning forest F of $K[B]$, and while F has more than one component, choose a vertex $a \in A$ not yet added that is adjacent to two components of F, and add a together with one edge to each component it meets. Such a vertex exists. Indeed, at every stage the vertices outside F lie in A and have all their neighbours in $B \subseteq V(F)$, and no edge of K joins two components of F (at the start because F spans $K[B]$, later because each added vertex is joined to every component it meets). So if every vertex outside F met only one component, then for

a component C of F the set $V(C)$, together with the vertices outside F adjacent to C, would be closed under adjacency (as A is independent) and different from $V(K)$, contradicting the connectivity of K. Each step reduces the number of components, so at most $\kappa-1$ vertices of A are added; let S be the set of them and T_0 the final tree. Every vertex of S has degree at least two in T_0, so the leaves of T_0 lie in B, and as $|V(T_0)| \geq |B| \geq 3$ there are two of them, b_0 and b_1. Attach every remaining vertex of A to one of its three neighbours (all in B) other than b_0, b_1. In the resulting spanning tree of K all internal vertices lie in $(B \setminus \{b_0, b_1\}) \cup S$; the internal vertices of a spanning tree with at least three vertices form a connected dominating set, so $\gamma_c(K) \leq |B|-2+\kappa-1 \leq 2|B|-W_B-3 = 2M(K)-3$. Finally $\rho(K) \leq (\gamma_c(K)+1)/2 \leq M(K)-1$ by Lemma 2.2. □

8.3. **Why bounded-degree arguments do not extend.**

Example 8.3. For $b \geq 2$ let N_b be $K_{b+1,b}$ together with a vertex h adjacent to all $2b+1$ vertices, and let G_b be N_b with a leaf at every vertex. Let X and Y be the parts of $K_{b+1,b}$ of sizes $b+1$ and b; in G_b the vertices of X, of Y and h have degrees $b+2$, $b+3$ and $2b+2$. By Lemma 3.2, deleting an X–Y, an X–h or a Y–h edge increases M by $\frac{2}{(b+1)(b+2)(b+3)}$, $\frac{1}{(b+1)(b+2)}$ and $\frac{1}{b+3}$ respectively, so G_b satisfies (a); (b) and (c) are clear. Moreover $M(G_b) = 1 + \frac{b+1}{b+2} + \frac{b(b+2)}{b+3}$ and $M(N_b) = 1 + \frac{b(b+1)}{b+2}$, so $M(G_b) - M(N_b) = \frac{b^2+5b+3}{b^2+5b+6} < 1$. Thus conditions (a)–(c), together with (d) for $L' = L$, do not by themselves bound $|K|$. On the other hand $\rho(N_b) = 1$ and $\rho(G_b) = \rho(G_b - \ell) = 2$ for every leaf ℓ, so no leaf of G_b is ρ-critical: G_b violates (e) (equivalently, (d) for $L' = \{\ell\}$), and indeed $M(G_b) - \rho(G_b) \to \infty$. Any proof of Conjecture 8.1 along the lines of Section 6 must therefore use (e), or (d) for sets $L' \neq L$; these conditions are metric rather than local.

Every minimal counterexample satisfies (a)–(e) of Proposition 5.3. By exhaustive search, there are 44 710 decorated cores with $|K| \leq 10$ that satisfy (a)–(c) and have at most two subs (41 627 of them with $|K| = 10$); all satisfy $M(G)-\rho(G) \geq \frac{5}{12}$. Only 23 of them satisfy $M(G)-M(K) < 1$, which (d) requires for $L' = L$, and only one of them has all its leaves ρ-critical (its core has 10 vertices and 25 edges, the degrees in G are $5,5,5,5,5,7,7,7,7,7$, and every vertex carries a leaf); for that one $M(G) - M(K) = \frac{47}{42} > 1$, so it violates (d). Proving that (a)–(e) are incompatible in general would prove Conjecture 1.1.

8.4. **A false strengthening.** One might hope that a vertex of maximum degree is always a good centre. This is true for all cores with at most 10 vertices: for every vertex h of maximum degree, $\min\{\mathrm{ecc}(h),\ \min_{w\sim h} \mathrm{ecc}(\text{midpoint of } hw)\} \leq M - 1 + \frac{s}{3}$. It is false in general.

Example 8.4. For $U \geq 2$ let C_U have vertices a_i, b_i $(0 \leq i \leq U)$ and y_i, y_i', z_i, z_i' $(0 \leq i < U)$, with a_i, b_i adjacent to y_i, y_i', with z_i, z_i' adjacent to a_{i+1}, b_{i+1}, with edges y_iz_i and $y_i'z_i'$, and with the edges a_0b_0 and a_Ub_U. Then C_U is 2-connected with $6U+2$ vertices, all degrees are 3 or 4, $s = 0$, the vertices of maximum degree are a_i, b_i $(1 \leq i \leq U-1)$, and $M(C_U) = \frac{8U+4}{3}$ (the $8(U-1)$ edges at vertices of degree 4 have weight $\frac{1}{4}$, the other $2U+10$ edges weight $\frac{1}{3}$). Every path from $\{a_i, b_i\}$ to $\{a_{i+1}, b_{i+1}\}$ has length at least 3, and $d(a_i, a_{i+1}) = d(a_i, b_{i+1}) = 3$.

Let $U \geq 3$ and $h = b_{U-1}$. Then $\mathrm{ecc}(h) = d(h, a_0) = d(h, b_0) = 3U-3$, and all vertices with index at least $U-1$ are within distance 3 of h. The neighbours of h are z_{U-2}, z_{U-2}' and y_{U-1}, y_{U-1}'. By Lemma 2.1(1) the midpoint of hz_{U-2} has eccentricity $\frac{1}{2} + d(z_{U-2}, a_0) = \frac{1}{2} + (3U-4)$, since $3U-4 \geq 3$; the midpoint of hy_{U-1} has eccentricity $\frac{1}{2} + d(h, a_0) = 3U - \frac{5}{2}$, and the edges hz_{U-2}', hy_{U-1}' are symmetric to these. Hence

$$\min\{\mathrm{ecc}(h),\ \min_{w\sim h} \mathrm{ecc}(\text{midpoint of } hw)\} = 3U - \tfrac{7}{2},$$

and the rooted inequality has slack $\frac{8U+1}{3} - \left(3U - \frac{7}{2}\right) = \frac{23-2U}{6}$, which is negative for every $U \geq 12$ (for $U = 12$: $n = 74$, $M - 1 = \frac{97}{3} < \frac{65}{2}$). On the other hand $d(a_0, a_U) = 3U$, and the eccentricity $\frac{3U}{2}$ is attained at $a_{U/2}$ if U is even and at the midpoint of $y_k z_k$, $k = \frac{U-1}{2}$, if U is odd; so $\rho(C_U) = \frac{3U}{2}$, and the core inequality holds with slack $\frac{7U+2}{6}$. We confirmed all these values by computer for $2 \leq U \leq 20$ (the rooted value for $3 \leq U \leq 20$).

Thus centres must be chosen globally, as in the splitting Lemma 4.3; a proof of Conjecture 8.1 by a breadth-first search from a vertex of maximum degree cannot work.

9. Computations

All programs use exact arithmetic: edge weights are scaled by a common multiple of $1, \ldots, \Delta$, and 2ρ is computed exactly from Lemma 2.1(1)–(2) by breadth-first search. Graphs were generated with `geng` from `nauty` [11]. The programs are listed in Appendix A; an independent Python implementation (using `networkx` and `fractions`) agrees on all graphs with at most 8 vertices.

n	cores tested	min. slack of (2)
4	2	0
5	6	1/6
6	44	1/6
7	385	1/4
8	6 268	3/10
9	179 303	1/3
10	9 349 191	5/14
11	884 325 784	3/8
n, m	cores with $\mu \leq 8$	
11, 16*	2 114	7/6
11, 17*	42 268	5/6
11, 18*	329 471	1
12, 17	744	4/3
12, 18	46 952	1
12, 19	745 461	7/6
13, 19	27 127	7/6
13, 20	1 028 368	4/3
14, 20	6 904	4/3
14, 21	852 101	3/2
15, 22	387 630	5/3
16, 23	77 080	11/6

Table 1. Exhaustive verification of the core inequality (Proposition 7.1). The rows marked * are contained in the row $n = 11$; they were run separately and serve as a cross-check.

Other checks. (1) $R'(G) \geq \rho(G)$ for all 1 006 700 565 connected graphs with 11 vertices and all smaller connected graphs. (2) The splitting inequality of Lemma 4.3 for every cut vertex and every bipartition of the components, for all connected graphs with at most 9 vertices. (3) The configurations of Proposition 6.4 and Remark 6.5, by `cfgcheck.c`. All statements of Proposition 6.4 (the 128 configurations, the values of $M(G) - \rho(G)$, and the tests of (d) and (e))

were reproduced by a second, independently written Python program with exact fractions. As a further cross-check, the 374 configurations with $\Delta \le 5$ and $|K| = 11$ also satisfy $M(G) > \rho(G)$ and violate both (d) for $L' = L$ and (e). (4) Exhaustive enumeration of all decorated cores satisfying (a)–(c) of Proposition 5.3 with at most two subs and $|K| \le 10$ (`cfgcheck.c` with $D = 0$; counts $1, 0, 2, 46, 241, 2793, 41627$ for $|K| = 4, \dots, 10$): every one satisfies $M(G) > \rho(G)$; 23 of them satisfy $M(G) - M(K) < 1$, one satisfies (e), and none satisfies both. (Independently, by the proof of Theorem 5.4 the core of a minimal counterexample violates (2), so by Proposition 7.1 it has at least 12 vertices.)

10. Concluding remarks

The absolute radius turns Fajtlowicz's problem into the exact statement $R' \ge \rho$, which is tight for all paths with at least three vertices and is preserved by the natural surgeries: splitting at cut vertices is exact for R' (Lemma 3.3) and sub-additive for ρ (Lemma 4.3). This reduces everything to 2-connected cores with pendant vertices, and with bounded degree the local constraints are strong enough to finish (Theorems 6.1 and 6.3). For unbounded degree, high-degree hubs make the local constraints satisfiable (Example 8.3) and the problem becomes the core inequality, or equivalently the incompatibility of conditions (d) and (e) of Proposition 5.3. We expect that a proof will need a global argument that combines the weighted max-leaf inequality (4) for hub-dominated cores (Proposition 8.2) with the metric condition (e).

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Appendix A. Programs

The programs read graphs in `graph6` format from standard input (as produced by `geng`) and use exact integer arithmetic. The invocations used for the results of the paper are as follows.

- Core inequality (Proposition 7.1, Table 1):
  `geng -C -d2 -q` n `| ./corechk` for $4 \le n \le 11$;
  `geng -C -d2 -q` n m`:`m `| ./corechk` for $(n, m) = (11, 16)$, $(11, 17)$, $(11, 18)$, $(12, 17)$, $(12, 18)$, $(12, 19)$, $(13, 19)$, $(13, 20)$, $(14, 20)$, $(14, 21)$;
  `geng -C -d2 -D4 -q 15 22:22 | ./corechk` and `geng -C -d2 -D3 -q 16 23:23 | ./corechk`.

- Configurations (Proposition 6.4, Remark 6.5, Section 9):
  `geng -C -d2 -D`D `-q` n `| ./cfgcheck` D for $D \in \{5, 6\}$ and $4 \le n \le 10$, and for $D = 5$, $n = 11$;
  `geng -C -d2 -q` n `| ./cfgcheck 0` for $4 \le n \le 10$.
- $R' \ge \rho$ for all connected graphs (Section 9, check (1)): `geng -c -q` n `| ./rabs` for $n \le 11$, the case $n = 11$ split into two parts with the `res/mod` option of `geng`.
- Domination inequalities of Section 8: `geng -c -d2 -q` n `| ./cscheck` for $n \le 10$.
- Independent check of $R' \ge \rho$ and of (2): `geng -c` n `| python3 independent_check.py` for $n \le 8$.

Check (2) of Section 9 (the splitting inequality of Lemma 4.3) used a direct implementation of the statement of that lemma, which is not listed here; Lemma 4.3 is proved in Section 4, and this check was only a sanity test.

corechk.c: the core inequality.

```
// Core inequality check: for each graph (graph6, stdin; assumed 2-connected, e.g. from geng -C) with min
     degree>=2 and
// s<=2 vertices of degree 2: verify rho(K) <= M(K) - 1 + s/3 exactly. W = lcm(1..maxdeg)*6 scaling.
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
typedef unsigned int u32;
static int n; static u32 A[32]; static int dist[32][32];
static void parse(const char*s){ n=s[0]-63; memset(A,0,sizeof A); const char*p=s+1; int bit=0,val=0;
  for(int j=1;j<n;j++) for(int i=0;i<j;i++){ if(bit==0){val=*p++-63; bit=6;} bit--; if(val>>bit&1){A[i
    ]|=1u<<j; A[j]|=1u<<i;} } }
static long long gcdll(long long a,long long b){ while(b){long long t=a%b;a=b;b=t;} return a; }
int main(){ char line[256]; long long cnt=0, viol=0; double minsl=1e9; char gmin[64]="";
  while(fgets(line,sizeof line,stdin)){ line[strcspn(line,"\n")]=0; if(!line[0]) continue; parse(line);
    int dg[32], s=0, bad=0, D=0; for(int i=0;i<n;i++){ dg[i]=__builtin_popcount(A[i]); if(dg[i]<2) bad=1;
     if(dg[i]==2) s++; if(dg[i]>D) D=dg[i]; }
    if(bad||s>2) continue; cnt++;
    long long W=6; for(int i=2;i<=D;i++) W=W/gcdll(W,i)*i;
    long long M=0; for(int u=0;u<n;u++) for(u32 a=A[u];a;a&=a-1){int v=__builtin_ctz(a); if(v>u) M+=W/(dg
     [u]>dg[v]?dg[u]:dg[v]);}
    for(int s0=0;s0<n;s0++){ for(int t=0;t<n;t++) dist[s0][t]=99; dist[s0][s0]=0; u32 seen=1u<<s0, fr=
     seen; int d=0; while(fr){ d++; u32 nf=0; for(u32 q=fr;q;q&=q-1) nf|=A[__builtin_ctz(q)]; nf&=~seen;
     for(u32 q=nf;q;q&=q-1) dist[s0][__builtin_ctz(q)]=d; seen|=nf; fr=nf; } }
    int r2=999; for(int u=0;u<n;u++){ int e=0; for(int t=0;t<n;t++) if(dist[u][t]>e) e=dist[u][t]; if(2*e
     <r2) r2=2*e; }
    for(int u=0;u<n;u++) for(u32 a=A[u];a;a&=a-1){ int v=__builtin_ctz(a); if(v<u) continue; int mx=0;
     for(int w=0;w<n;w++){ int q=dist[u][w]<dist[v][w]?dist[u][w]:dist[v][w]; if(q>mx) mx=q; } if(2*mx+1<
     r2) r2=2*mx+1; }
    // check  r2/2 <= M/W - 1 + s/3  <=>  3*W*r2 <= 6*M - 6*W + 2*W*s
    long long lhs=3*W*(long long)r2, rhs=6*M-6*W+2*W*(long long)s;
    double sl=(double)(rhs-lhs)/(6.0*W);
    if(sl<minsl){minsl=sl; strcpy(gmin,line);}
    if(lhs>rhs){ viol++; printf("VIOLATION %s\n",line); } }
  fprintf(stderr,"cores=%lld violations=%lld min slack=%.4f at %s\n",cnt,viol,minsl,gmin);
}
```

cfgcheck.c: decorated cores satisfying (a)–(c) of Proposition 5.3, and the tests of (d) and (e).

```
// Enumerate decorated cores G=K+L satisfying (a)-(c) of the structure proposition and at most two subs:
//  K 2-connected (input from geng -C -d2), one leaf per decorated vertex; vertices of K-degree 2 or 3
     are decorated
//  (forced by (c)); every K-vertex has a G-neighbour of smaller degree; subs (G-degree 3) pairwise non-
     adjacent, <=2;
//  every K-edge supportive. Optional Delta bound D (0 = none): then vertices of K-degree D carry no leaf
     .
// For every configuration it computes M(G)-rho(G) exactly and tests three necessary conditions of the
     proposition:
```

```
//  (dL)  M(G)-M(G-L) < 1                      [(d) with L'=L]
//  (d4)  L4 empty or M(G)-M(G-L4) < 1         [(d) with L'=L4 = leaves at hosts of G-degree <= 4]
//  (e)   every leaf is rho-critical: rho(G-l) < rho(G)
// Usage: geng -C -d2 [-D<D>] -q n | ./cfgcheck D [v]   (v: also print the configurations that pass (d4))
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
typedef unsigned int u32;
static int nk; static u32 A[32];
static long long W;
static void parseK(const char*s){ nk=s[0]-63; memset(A,0,sizeof A); const char*p=s+1; int bit=0,val=0;
  for(int j=1;j<nk;j++) for(int i=0;i<j;i++){ if(bit==0){val=*p++-63; bit=6;} bit--; if(val>>bit&1){A[i
     ]|=1u<<j; A[j]|=1u<<i;} } }
static long long Mint(u32*G,int m){ long long s=0; for(int u=0;u<m;u++) for(u32 a=G[u];a;a&=a-1){int v=
     __builtin_ctz(a); if(v>u){int du=__builtin_popcount(G[u]),dv=__builtin_popcount(G[v]); s+=W/(du>dv?
     du:dv);}} return s; }
static int dist[32][32];
static int rho2(u32*G,int m){ for(int s=0;s<m;s++){ for(int t=0;t<m;t++) dist[s][t]=99; dist[s][s]=0; u32
      seen=1u<<s, fr=seen; int d=0; while(fr){ d++; u32 nf=0; for(u32 q=fr;q;q&=q-1) nf|=G[__builtin_ctz(
     q)]; nf&=~seen; for(u32 q=nf;q;q&=q-1) dist[s][__builtin_ctz(q)]=d; seen|=nf; fr=nf; } }
  int best=999; for(int s=0;s<m;s++){ int e=0; for(int t=0;t<m;t++) if(dist[s][t]>e) e=dist[s][t]; if(2*e
     <best) best=2*e; }
  for(int u=0;u<m;u++) for(u32 a=G[u];a;a&=a-1){ int v=__builtin_ctz(a); if(v<u) continue; int mx=0; for(
     int w=0;w<m;w++){ int q=dist[u][w]<dist[v][w]?dist[u][w]:dist[v][w]; if(q>mx) mx=q; } if(2*mx+1<best
     ) best=2*mx+1; }
  return best; }
// induced subgraph of G (m vertices) on the vertex set keep (bitmask), relabelled 0..k-1; returns k
static int induced(u32*G,int m,u32 keep,u32*H){ int idx[32],k=0; for(int v=0;v<m;v++) if(keep>>v&1) idx[v
     ]=k++;
  memset(H,0,32*sizeof(u32)); for(int v=0;v<m;v++) if(keep>>v&1) for(u32 a=G[v]&keep;a;a&=a-1) H[idx[v
     ]]|=1u<<idx[__builtin_ctz(a)];
  return k; }
int main(int argc,char**argv){ int D=argc>1?atoi(argv[1]):0, verbose=argc>2; W=27720; // lcm(1..12)
  char line[256]; long long cores=0, confs=0, ndL=0, nd4=0, ne=0, ndLe=0, nall=0;
  double minsl=1e9, mindropc=1e9; char gmin[64]="", gdc[64]=""; u32 smin=0, sdc=0;
  while(fgets(line,sizeof line,stdin)){ line[strcspn(line,"\n")]=0; if(!line[0]) continue; parseK(line);
     cores++;
    int deg[32]; for(int i=0;i<nk;i++) deg[i]=__builtin_popcount(A[i]);
    u32 forced=0, freev=0; int ns0=0, bad=0;
    for(int i=0;i<nk;i++){ if(deg[i]<2){bad=1;break;} if(deg[i]<=3) forced|=1u<<i; else if(!(D && deg[i
     ]>=D)) freev|=1u<<i; if(deg[i]==2) ns0++; }
    if(bad||ns0>2) continue;
    if(D) { int ok=1; for(int i=0;i<nk;i++) if(deg[i]>D) ok=0; if(!ok) continue; }
    u32 sub=0; do { u32 S=forced|sub; int dg[32]; for(int i=0;i<nk;i++) dg[i]=deg[i]+((S>>i)&1);
      int ok=1; u32 subs=0; for(int i=0;i<nk;i++) if(dg[i]==3) subs|=1u<<i;
      if(__builtin_popcount(subs)>2) ok=0;
      for(u32 q=subs;q&&ok;q&=q-1){ int i=__builtin_ctz(q); if(A[i]&subs){ok=0;break;} }
      if(ok) for(int i=0;i<nk && ok;i++) if(!((S>>i)&1)){ int has=0; for(u32 a=A[i];a;a&=a-1) if(dg[
     __builtin_ctz(a)]<dg[i]){has=1;break;} if(!has) ok=0; }
      if(ok){ for(int u=0;u<nk && ok;u++) for(u32 a=A[u];a;a&=a-1){ int v=__builtin_ctz(a); if(v<u)
     continue;
           int lu=((S>>u)&1), lv=((S>>v)&1);
           for(u32 b=A[u]&~(1u<<v);b;b&=b-1){ int w=__builtin_ctz(b); if(dg[w]<=dg[u]-1) lu++; }
           for(u32 b=A[v]&~(1u<<u);b;b&=b-1){ int w=__builtin_ctz(b); if(dg[w]<=dg[v]-1) lv++; }
           int du=dg[u], dv=dg[v], mx=du>dv?du:dv;
           long long L=(long long)lu*dv*(dv-1)*mx + (long long)lv*du*(du-1)*mx; long long R=(long long)du
     *(du-1)*dv*(dv-1);
           if(!(L>R)){ok=0;break;} } }
      if(ok){ confs++;
        u32 G[32]; memset(G,0,sizeof G); memcpy(G,A,sizeof A); int n=nk; int host[32];
        for(int i=0;i<nk;i++) if(S>>i&1){ G[i]|=1u<<n; G[n]=1u<<i; host[n]=i; n++; }
        long long Mi=Mint(G,n), MK=Mint(A,nk); int r2=rho2(G,n); double sl=(double)Mi/W - r2/2.0;
        if(sl<minsl){minsl=sl; strcpy(gmin,line); smin=S;}
```

```
        if(sl<0) printf("COUNTEREXAMPLE %s S=%x\n",line,S);
        int okL=(Mi-MK)<W; if(okL) ndL++;
        u32 all=(n==32)?0xffffffffu:((1u<<n)-1), keep=all; for(int l=nk;l<n;l++) if(dg[host[l]]<=4) keep
    &=~(1u<<l);
        int ok4=1; if(keep!=all){ u32 H[32]; int k=induced(G,n,keep,H); ok4=(Mi-Mint(H,k))<W; } if(ok4)
    nd4++;
        int allc=1; for(int l=nk;l<n && allc;l++){ u32 H[32]; int k=induced(G,n,all&~(1u<<l),H); if(!(
    rho2(H,k)<r2)) allc=0; }
        if(allc){ ne++; double dr=(double)(Mi-MK)/W; if(dr<mindropc){mindropc=dr; strcpy(gdc,line); sdc=S
    ;} }
        if(allc&&okL) { ndLe++; printf("PASS(dL,e) %s S=%x\n",line,S); }
        if(allc&&okL&&ok4) { nall++; printf("PASS(dL,d4,e) %s S=%x\n",line,S); }
        if(verbose&&ok4) printf("PASS(d4) %s S=%x s=%d M-rho=%.4f all-critical=%d\n",line,S,
    __builtin_popcount(subs),sl,allc);
      }
      sub=(sub-freev)&freev; } while(sub);
  }
  fprintf(stderr,"D=%d cores=%lld configs=%lld min(M-rho)=%.4f [%s S=%x] pass(dL):%lld pass(d4):%lld pass
    (e):%lld "
                "(min drop(L) among pass(e): %.4f [%s S=%x]) pass(dL,e):%lld pass(all):%lld\n",
          D,cores,confs,minsl,gmin,smin,ndL,nd4,ne,mindropc,gdc,sdc,ndLe,nall);
}
```

rabs.c: **the inequality $R' \geq \rho$ for all connected graphs.**

```
// Strong-M test: M(G) >= r_abs(G), where r_abs = r-1/2 if some edge uv has max_w min(d(u,w),d(v,w)) <= r
    -1, else r.
// Equivalently: G has a spanning tree of diameter <= 2M(G).
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
typedef unsigned int u32;
static int n; static u32 adj[32]; static long long W;
static long long lcm(long long a,long long b){long long x=a,y=b; while(y){long long t=x%y;x=y;y=t;}
    return a/x*b;}
static void parse(const char*s){ n=s[0]-63; memset(adj,0,sizeof adj); const char*p=s+1; int bit=0,val=0;
  for(int j=1;j<n;j++) for(int i=0;i<j;i++){ if(bit==0){val=*p++-63; bit=6;} bit--; if(val>>bit&1){adj[i
    ]|=1u<<j; adj[j]|=1u<<i;} } }
static int dist[32][32];
static void apsp(){ u32 all=(1u<<n)-1; for(int s=0;s<n;s++){ for(int t=0;t<n;t++) dist[s][t]=99; u32 seen
    =1u<<s, fr=1u<<s; int e=0; dist[s][s]=0;
  while(fr){u32 nf=0; for(u32 f=fr;f;f&=f-1) nf|=adj[__builtin_ctz(f)]; nf&=all&~seen; if(!nf)break; e++;
     for(u32 g=nf;g;g&=g-1) dist[s][__builtin_ctz(g)]=e; seen|=nf; fr=nf;} } }
int main(int argc,char**argv){
  char line[256]; long long cnt=0, viol=0, tightE=0, tightV=0, below=0; double minsl=1e9; char mins
    [256]="";
  while(fgets(line,sizeof line,stdin)){
    line[strcspn(line,"\n")]=0; parse(line); cnt++; if(n<2) continue;
    W=1; for(int i=2;i<n;i++) W=lcm(W,i);
    apsp(); int r=99; for(int s=0;s<n;s++){int e=0; for(int t=0;t<n;t++) if(dist[s][t]>e) e=dist[s][t];
     if(e<r) r=e;}
    long long Mi=0; for(int u=0;u<n;u++) for(u32 a=adj[u];a;a&=a-1){int v=__builtin_ctz(a); if(v>u){int
     du=__builtin_popcount(adj[u]),dv=__builtin_popcount(adj[v]); Mi+=W/(du>dv?du:dv);}}
    int edgec=0;
    for(int u=0;u<n&&!edgec;u++) for(u32 a=adj[u];a;a&=a-1){int v=__builtin_ctz(a); if(v<u) continue; int
      mx=0; for(int w=0;w<n;w++){int d=dist[u][w]<dist[v][w]?dist[u][w]:dist[v][w]; if(d>mx) mx=d;} if(mx
     <=r-1){edgec=1;break;} }
    // 2*r_abs = 2r-1 if edgec else 2r ; test 2M >= 2 r_abs
    long long lhs=2*Mi, rhs=W*(edgec?(2*r-1):(2*r));
    double sl=(double)(lhs-rhs)/W/2.0;
    if(sl<minsl){minsl=sl; strcpy(mins,line);}
    if(lhs<rhs){viol++; if(viol<=50) printf("VIOL %s n=%d r=%d edgec=%d M=%.6f\n",line,n,r,edgec,(double)
     Mi/W);}
    else if(lhs==rhs){ if(edgec) tightE++; else tightV++; }
```

```
    if(2*Mi < 2*W*r) below++;
  }
  fprintf(stderr,"graphs=%lld violations=%lld tight(edge-centered)=%lld tight(vertex-centered)=%lld M<r:%
     lld minslack=%.6f at %s\n",cnt,viol,tightE,tightV,below,minsl,mins);
}
```

cscheck.c: **the domination inequalities of Section 8.**

```
// Exact check of (Cs): gamma_c(G) <= 2M(G) - max(2, 3 - s/3) for connected graphs with min degree >= 2 (
     graph6 stdin),
// and of (LG'): gamma_c <= 2M - 3 + 2s/3 for cores (2-connected, s<=2). Integer arithmetic with W = 3*
     lcm(1..n).
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
typedef unsigned int u32;
static int n; static u32 adj[32];
static void parse(const char*s){ n=s[0]-63; memset(adj,0,sizeof adj); const char*p=s+1; int bit=0,val=0;
  for(int j=1;j<n;j++) for(int i=0;i<j;i++){ if(bit==0){val=*p++-63; bit=6;} bit--; if(val>>bit&1){adj[i
     ]|=1u<<j; adj[j]|=1u<<i;} } }
static int conn_set(u32 S){ if(!S) return 0; u32 seen=S&-S, fr=seen; while(fr){ u32 nf=0; for(u32 q=fr;q;
     q&=q-1) nf|=adj[__builtin_ctz(q)]; nf&=S&~seen; seen|=nf; fr=nf;} return seen==S; }
static int gammac(void){ u32 all=(1u<<n)-1; for(int k=1;k<=n;k++){ u32 S=(1u<<k)-1; while(S<=all){ u32
     dom=S; for(u32 q=S;q;q&=q-1) dom|=adj[__builtin_ctz(q)]; if(dom==all && conn_set(S)) return k;
      u32 c=S&-S, r=S+c; S=(((r^S)>>2)/c)|r; if(S==0) break; } } return n; }
int main(){ char line[256]; long long cnt=0, cores=0, vcs=0, vlg=0; double mcs=1e9, mlg=1e9; char gcs
     [64]="", glg[64]="";
  long long W=3; for(int i=2;i<=12;i++){ long long a=W,b=i; while(b){long long t=a%b;a=b;b=t;} W=W/a*i; }
  while(fgets(line,sizeof line,stdin)){ line[strcspn(line,"\n")]=0; if(!line[0]) continue; parse(line);
    int dg[32], s=0, bad=0; for(int i=0;i<n;i++){ dg[i]=__builtin_popcount(adj[i]); if(dg[i]<2) bad=1; if
     (dg[i]==2) s++; } if(bad||n<3) continue; cnt++;
    long long M2=0; for(int u=0;u<n;u++) for(u32 a=adj[u];a;a&=a-1){int v=__builtin_ctz(a); if(v>u) M2
     +=2*W/(dg[u]>dg[v]?dg[u]:dg[v]);}
    int g=gammac();
    long long t = (3*W - s*W/3 > 2*W) ? 3*W - s*W/3 : 2*W;   // W*max(2,3-s/3)
    double sl=(double)(M2 - t - (long long)g*W)/W; if(sl<mcs){mcs=sl; strcpy(gcs,line);} if(sl<-1e-12){
     vcs++; printf("Cs-VIOLATION %s\n",line);}
    if(s<=2){ u32 all=(1u<<n)-1; int tc=1; for(int x=0;x<n;x++) if(!conn_set(all&~(1u<<x))){tc=0;break;}
      if(tc){ cores++; double sl2=(double)(M2 - 3*W + 2*s*W/3 - (long long)g*W)/W; if(sl2<mlg){mlg=sl2;
     strcpy(glg,line);} if(sl2<-1e-12){vlg++; printf("LG-VIOLATION %s\n",line);} } } }
  fprintf(stderr,"leafless=%lld (Cs) violations=%lld min slack=%.4f [%s] | cores=%lld (LG') violations=%
     lld min slack=%.4f [%s]\n",cnt,vcs,mcs,gcs,cores,vlg,mlg,glg);
}
```

independent_check.py: **independent check in Python.**

```
#!/usr/bin/env python3
"""Independent check (pure Python + networkx, exact fractions) of
   (1) R'(G) >= rho(G) for all connected graphs on n vertices (graph6 on stdin),
   (2) the core inequality rho(K) <= R'(K) - 1 + s/3 for 2-connected K with s <= 2 degree-2 vertices.
The absolute radius is computed directly from its definition on the metric graph:
   rho = min( min_v ecc(v), min_{uv in E} (1/2 + max_w min(d(u,w), d(v,w))) ).
Usage:  nauty-geng -c 7 | python3 independent_check.py
"""
import sys
from fractions import Fraction as F
import networkx as nx

def rprime(G):
    d = dict(G.degree())
    return sum(F(1, max(d[u], d[v])) for u, v in G.edges())

def abs_radius(G):
    dist = dict(nx.all_pairs_shortest_path_length(G))
    best = min(F(max(dist[v].values())) for v in G)
```

```
    for u, v in G.edges():
        val = F(1, 2) + max(min(dist[u][w], dist[v][w]) for w in G)
        best = min(best, val)
    return best

def main():
    count = viol = core_count = core_viol = 0
    worst = None
    for line in sys.stdin:
        s = line.strip()
        if not s:
            continue
        G = nx.from_graph6_bytes(s.encode())
        if G.number_of_nodes() < 2:
            continue
        count += 1
        M, rho = rprime(G), abs_radius(G)
        if M < rho:
            viol += 1
            print("VIOLATION of R' >= rho:", s, M, rho)
        if worst is None or M - rho < worst[0]:
            worst = (M - rho, s)
        deg = dict(G.degree())
        s2 = sum(1 for v in G if deg[v] == 2)
        if G.number_of_nodes() >= 3 and min(deg.values()) >= 2 and s2 <= 2 and nx.is_biconnected(G):
            core_count += 1
            if rho > M - 1 + F(s2, 3):
                core_viol += 1
                print("VIOLATION of core inequality:", s, M, rho, s2)
    print(f"graphs={count} violations={viol} min(R'-rho)={worst[0] if worst else None} at {worst[1] if
     worst else None}")
    print(f"cores tested={core_count} core violations={core_viol}")

if __name__ == "__main__":
    main()
```